Reference guide: Python concepts from module 3

Google Cybersecurity Certificate

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Built-in functions

The following built-in functions are commonly used in Python.

`str()`

Converts the input object to a string

> `str(10)`
> Converts the integer `10` to the string `"10"`

`len()`

Returns the number of elements in an object

```
print(len("security"))
```

Returns and displays `8`, the number of characters in the string `"security"`

String methods

The following methods can be applied to strings in Python.

`.upper()`

Returns a copy of the string in all uppercase letters

```
print("Security".upper())
```

Returns and displays a copy of the string `"Security"` as `"SECURITY"`

`.lower()`

Returns a copy of the string in all lowercase letters

```
print("Security".lower())
```

Returns and displays a copy of the string `"Security"` as `"security"`

`.index()`

Finds the first occurrence of the input in a string and returns its location

```
print("Security".index("c"))
```

Finds the first occurrence of the character `"c"` in the string `"Security"` and returns and displays its index of `2`

List methods

The following methods can be applied to lists in Python.

`.insert()`

Adds an element in a specific position inside the list

```
username_list = ["elarson", "fgarcia", "tshah"]
username_list.insert(2,"wjaffrey")
```

Adds the element `"wjaffrey"` at index `2` to the `username_list`; the list becomes `["elarson", "fgarcia", "wjaffrey", "tshah"]`

`.remove()`

Removes the first occurrence of a specific element inside a list

```
username_list = ["elarson", "bmoreno", "wjaffrey", "tshah"]
username_list.remove("elarson")
```

Removes the element `"elarson"` from the `username_list`; the list becomes `["fgarcia", "wjaffrey", "tshah"]`

`.append()`

Adds input to the end of a list

```
username_list = ["bmoreno", "wjaffrey", "tshah"]
username_list.append("btang")
```

Adds the element `"btang"` to the end of the `username_list`; the list becomes `["fgarcia", "wjaffrey", "tshah", "btang"]`

`.index()`

Finds the first occurrence of an element in a list and returns its index

```
username_list = ["bmoreno", "wjaffrey", "tshah", "btang"]
print(username_list.index("tshah"))
```

Finds the first occurrence of the element `"tshah"` in the `username_list` and returns and displays its index of `2`

Additional syntax for working with strings and lists

The following syntax is useful when working with strings and lists.

+ (concatenation)

Combines two strings or lists together

> `device_id = "IT"+"nwp12"`
> Combines the string `"IT"` with the string `"nwp12"` and assigns the combined string of `"ITnwp12"` to the variable `device_id`
>
> `users = ["elarson", "bmoreno"] + ["tshah", "btang"]`
> Combines the list `["elarson", "bmoreno"]` with the list `["tshah", "btang"]` and assigns the combined list of `["elarson", "bmoreno", "tshah", "btang"]` to the variable `users`

[] (bracket notation)

Uses indices to extract parts of a string or list

> `print("h32rb17"[0])`
> Extracts the character at index `0`, which is (`"h"`), from the string `"h32rb17"`
>
> `print("h32rb17"[0:3])`
> Extracts the slice `[0:3]`, which is (`"h32"`), from the string `"h32rb17"`; the first index in the slice (`0`) is included in the slice but the second index in the slice (`3`) is excluded
>
> ```
> username_list = ["elarson", "fgarcia", "tshah"]
> print(username_list[2])
> ```
> Extracts the element at index `2`, which is (`"tshah"`), from the `username_list`

Regular expressions

The following `re` module function and regular expression symbols are useful when searching for patterns in strings.

`re.findall()`

Returns a list of matches to a regular expression

```
import re
re.findall("a53", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"a53"` in the string `"a53-32c .E"`; returns the list `["a53"]`

`\w`

Matches with any alphanumeric character; also matches with the underscore (_)

```
import re
re.findall("\w", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\w"` in the string `"a53-32c .E"`; matches to any alphanumeric character and returns the list `["a", "5", "3", "3", "2", "c", "E"]`

`.`

Matches to all characters, including symbols

```
import re
re.findall(".", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"."` in the string `"a53-32c .E"`; matches to all characters and returns the list `["a", "5", "3", "-", "3", "2", "c", " ", ".", "E"]`

`\d`

Matches to all single digits

```
import re
re.findall("\d", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\d"` in the string `"a53-32c .E"`; matches to all single digits and returns the list `["5", "3", "3", "2"]`

`\s`

Matches to all single spaces

```
import re
re.findall("\d", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\s"` in the string `"a53-32c .E"`; matches to all single spaces and returns the list `[" "]`

`\.`

Matches to the period character

```
import re
re.findall("\.", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\."` in the string `"a53-32c .E"`; matches to all instances of the period character and returns the list `["."]`

`+`

Represents one or more occurrences of a specific character

```
import re
re.findall("\w+", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\w+"` in the string `"a53-32c .E"`; matches to one or more occurrences of any alphanumeric character and returns the list `["a53", "32c", "E"]`

*****

Represents, zero, one or more occurrences of a specific character

```
import re
re.findall("\w*", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\w*"` in the string `"a53-32c .E"`; matches to one or more occurrences of any alphanumeric character and returns the list `["a53", " ", "32c", " ", " ", "E"]`

{ }

Represents a specified number of occurrences of a specific character; the number is specified within the curly brackets

```
import re
re.findall("\w{3}", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern `"\w{3}"` in the string `"a53-32c .E"`; matches to exactly three occurrences of any alphanumeric character and returns the list `["a53","32c"]`